A WINDOW–NIKOLSKII INEQUALITY: SHARP ONE-SIDED BOUNDS FOR LITTLEWOOD COSINE POLYNOMIALS AND DENSE CHOWLA SETS

DRAFT — CHOWLA COSINE PROJECT

ABSTRACT. Let $T(x) = \sum_{j=1}^{n} \mathrm{Re}(b_j e^{ijx})$ with $|b_j| \le 1$. We prove a window–Nikolskii inequality relating $M = -\min_{x\in\mathbb{R}} T(x)$ to the coefficient energy $E = \sum |b_j|^2$ and to the energy in low frequencies. Two consequences follow. (i) For every choice of phases, $-\min_x \sum_{j=1}^{n} \cos(jx - \theta_j) \ge 0.28\sqrt{n}$, and the same lower bound holds for the maximum; in particular every Littlewood cosine polynomial $\sum_{j=0}^{n} a_j \cos jx$, $a_j = \pm 1$, has $-\min \ge 0.28\sqrt{n} - 1$. This is the sharp order (Rudin–Shapiro), improving the recently claimed $n^{1/3}/60$. (ii) Chowla's cosine conjecture holds for dense sets: if $A \subset \mathbb{N}$ is finite with $|A| \ge 0.41 \max A$, then $-\min_x \sum_{a\in A} \cos ax \ge c\sqrt{|A|}$ with an absolute $c > 0$; this is the sharp order and improves the $|A|^{1/4}$ bound recently obtained for $A \subset [1, C|A|]$ in that density range. The same inequality gives $\sqrt{|A|}$ whenever the frequency profile of A is not front-loaded. We explain why the method cannot be localized in frequency (it is exactly critical on locally full sets) and show that the excluded front-loaded class contains sets with $-\min f_A/\sqrt{|A|} \to 1/2$. The 0/1 case of the main inequality and the dense theorem are formalized in Lean 4 (kernel-checked, standard axioms only).

1. Introduction

For a finite set A of positive integers let $f_A(x) = \sum_{a\in A} \cos(ax)$ and $M(A) = -\min_{x\in\mathbb{R}} f_A(x)$. Chowla conjectured $M(A) \ge c\sqrt{|A|}$. The best general lower bound is polynomial of exponent $1/5$ (Bedert [1], Shankar [4]); the best constructions give $M(A) \le (\frac{1}{2} + o(1))\sqrt{|A|}$ (Bu et al. [2]). For dense sets, $A \subset [1, C|A|]$, Bu et al. prove $M(A) \gg_C |A|^{1/4}$ and remark that this says nothing about whether $\sqrt{|A|}$ is the true order [2, Prop. 9.7, Rem. 9.8].

The one-sided Littlewood problem asks the same question for unimodular coefficients: for $T_n(x) = \sum_{j=1}^{n} \cos(jx - \theta_j)$, how small can $-\min T_n$ be? Since $\frac{1}{2\pi}\int_0^{2\pi} T_n = 0$ and $\frac{1}{2\pi}\int_0^{2\pi} T_n^2 = n/2$, one has $\max(\max T_n, -\min T_n) \ge \sqrt{n}/2$ trivially, but each one-sided quantity separately is harder; Erdélyi [3] recently claimed $-\min T_n \ge n^{1/3}/60$ and $\max T_n \ge n^{1/3}/60$.

Both questions are instances of one inequality.

2. The window–Nikolskii inequality

For $w \in \mathbb{N}$ let $\kappa_k = (1 - |k|/(w+1))_+$ and $K_w(y) = \sum_{|k|\le w} \kappa_k e^{iky} \ge 0$ (Fejér kernel). Put

$$S_w = \sum_{j=1}^{w} \kappa_j^2 = \frac{w(2w+1)}{6(w+1)}, \qquad W_w = \sum_{|k|\le w} \kappa_k^2 = 1 + 2S_w.$$

For $L \in \mathbb{N}$ let C_L be any constant such that

(1)
$$\frac{1}{2\pi}\int_0^{2\pi} h^2 \le C_L\Big(\frac{1}{2\pi}\int_0^{2\pi} h\Big)^2 \quad \text{for every nonnegative trigonometric polynomial } h \text{ of degree} \le L.$$

Lemma 1 (Nikolskii constants)**.** (1) *holds with* $C_L = L + 1$ *and with* $C_L = 0.811(L+1) + 71/300$.

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Proof. By Fejér–Riesz, $h = |U|^2$ with $U = \sum_{k=0}^{L} u_k e^{ikx}$. Then $h \le (\sum |u_k|)^2 \le (L+1)\sum |u_k|^2 = (L+1)\frac{1}{2\pi}\int_0^{2\pi} h$, and $\frac{1}{2\pi}\int_0^{2\pi} h^2 \le \max h \cdot \frac{1}{2\pi}\int_0^{2\pi} h$. The second constant is the bound $\sum_m |(u * \tilde{u})(m)|^2 \le (0.811(L+1)+71/300)(\sum |u_k|^2)^2$ for the autocorrelation of a sequence of length $L+1$; its elementary proof (a cumulative-mass estimate) is formalized as `elementary_analyticNikolskii`. □

Theorem 2. *Let $b_1, \dots, b_n \in \mathbb{C}$, $E = \sum_j |b_j|^2$, $T(x) = \sum_{j=1}^{n} \mathrm{Re}(b_j e^{ijx})$, and suppose $T \ge -M$ on $\mathbb{R}$. Then for every integer $0 \le w \le n$, with $L = n + w$,*

$$W_w\Big(M^2 + \frac{E}{2}\Big) \;\le\; C_L\Big(M^2 + \frac{1}{2}\sum_{j=1}^{w} \kappa_j^2 |b_j|^2\Big).$$

Proof. Let $g = T + M \ge 0$. For $x_0 \in \mathbb{R}$ the function $h_{x_0}(x) = K_w(x - x_0)g(x)$ is a nonnegative trigonometric polynomial of degree $\le L$, so $\frac{1}{2\pi}\int_0^{2\pi} h_{x_0}^2 \le C_L I(x_0)^2$ with $I(x_0) = \frac{1}{2\pi}\int_0^{2\pi} h_{x_0}$. Since $\hat{g}(0) = M$ and $\hat{g}(\pm j) = b_j/2, \bar{b}_j/2$,

$$I(x_0) = M + \sum_{j=1}^{w} \kappa_j \,\mathrm{Re}(b_j e^{ijx_0}), \qquad \frac{1}{2\pi}\int_0^{2\pi} I(x_0)^2\, dx_0 = M^2 + \frac{1}{2}\sum_{j\le w} \kappa_j^2 |b_j|^2.$$

On the other hand, by Fubini and $\frac{1}{2\pi}\int_0^{2\pi} K_w(x-x_0)^2 dx_0 = W_w$ for every x, $\frac{1}{2\pi}\int_0^{2\pi}\frac{1}{2\pi}\int_0^{2\pi} h_{x_0}(x)^2\, dx\, dx_0 = W_w \frac{1}{2\pi}\int_0^{2\pi} g^2 = W_w(M^2 + E/2)$. Averaging the pointwise inequality over x_0 gives the claim. □

Only the moduli $|b_j|$ enter: the inequality is blind to phases.

3. Unimodular and Littlewood cosine polynomials

Corollary 3. *For every $n \ge 1$ and all $\theta_1, \dots, \theta_n \in \mathbb{R}$,*

$$-\min_{x\in\mathbb{R}} \sum_{j=1}^{n} \cos(jx - \theta_j) \;\ge\; 0.28\sqrt{n}, \qquad \max_{x\in\mathbb{R}} \sum_{j=1}^{n} \cos(jx - \theta_j) \;\ge\; 0.28\sqrt{n}.$$

With the trivial constant $C_L = L+1$ the bound is $\ge 0.189\sqrt{n}$, with asymptotic constant $\sqrt{7/2 - 2\sqrt{3}}$. Consequently every $\sum_{j=0}^{n} a_j \cos jx$ with $a_j \in \{-1, 1\}$ satisfies $-\min \ge 0.28\sqrt{n} - 1$ and $\max \ge 0.28\sqrt{n} - 1$.

Proof. Take $|b_j| = 1$, so $E = n$ and $\sum_{j\le w} \kappa_j^2 |b_j|^2 = S_w$. With $C_L = L + 1$ and $W_w = 1 + 2S_w$, Theorem 2 rearranges to

$$M^2 \;\ge\; \frac{n + S_w(n - w - 1)}{2(n + w - 2S_w)} \qquad (0 \le w \le n - 1).$$

Using $w/3 - 1/6 \le S_w \le w/3$ and $w = \lfloor \theta n \rfloor$ gives $M^2 \ge n\,\theta(1-\theta)/(6 + 2\theta) - O(1)$, maximized at $\theta = 2\sqrt{3} - 3$ with value $(7/2 - 2\sqrt{3})n$. With the second constant of Lemma 1 the same computation (optimum $w \approx 0.69n$) gives $M \ge 0.2819\sqrt{n} - o(\sqrt{n})$; the exact bound, evaluated for every $n \le 2000$ and estimated beyond, stays above $0.28\sqrt{n}$. The maximum follows by applying this to $-T$ (phases $\theta_j + \pi$), and the ± 1 statement by adding the constant term a_0. □

Remark 4. The order is sharp: for $n = 2^k$ the Rudin–Shapiro signs give $|\sum_{j<n} \varepsilon_j e^{ijx}| \le \sqrt{2n}$, hence $|\sum_{j=1}^{n} \varepsilon_{j-1} \cos jx| \le \sqrt{2n}$. Numerically, optimizing the phases gives $-\min T_n/\sqrt{n} \approx 0.607$ for $8 \le n \le 384$; chirp phases give ≈ 1.17 and random phases ≈ 2.6.

4. Dense Chowla sets

Corollary 5. *If $A \subset \mathbb{N}$ is finite and nonempty with $|A| \geq \frac{41}{100} \max A$, then $M(A) \geq \sqrt{|A|}/1000$. More precisely, if $|A| \geq \delta \max A$ with $\delta > 0.4055$, then $M(A) \geq c(\delta)\sqrt{|A|}$ with, for large $|A|$, $c(0.5) \approx 0.064$ and $c(1) \approx 0.23$.*

Proof sketch. Apply Theorem 2 with $b_a = \mathbf{1}_A(a)$ (so $E = |A|$), $X = \max A$, and bound $\sum_{a \leq w} \kappa_a^2 \leq S_w \approx w/3$. For $w = \theta X$ the inequality becomes $\frac{2w}{3} \cdot \frac{|A|}{2} \lesssim 0.811(X + w)\frac{w}{6} + O(M^2 X)$, which forces $M^2 \gg w$ as soon as $|A|/X > 0.4055(1 + \theta)$. The explicit constants are computed in the formal proof. □

Remark 6 (Front-loading). More generally, Theorem 2 gives $M(A) \gg \sqrt{|A|}$ whenever the weighted low-frequency count $\sum_{a \in A}(1 - a/(w + 1))_+^2$ is at most $(1 - \eta)\frac{2}{3\rho}\frac{w}{X+w}|A|$ for some w, where $\rho = C_L/(L + 1)$. Sets violating this at every scale are *front-loaded*, and the class contains very good sets. Adding a few far elements to a Sidon set S keeps it Sidon, so Bu et al.'s identity $f_{A(S)} = \rho_S^2 - |S|/2 - g_S/2$ survives while the profile of $A(S)$ becomes arbitrarily front-loaded. Numerically (Bu's sets with $|S| = 40, 80$ plus 2–4 outliers, front-loading factors up to 200), $M(A)/\sqrt{|A|}$ stays within 0.03 of the value for $A(S)$ itself, e.g. 0.673 at $|A| = 6972$. In fact, writing $B = \max_x(g_S/2 - \rho_S^2)$ $(= O(\sqrt{k \log k})$ for Bu et al.'s sets, $k = |S|)$ and adding $k_o \leq \sqrt{k/2}$ outliers O, one has $g/2 - \rho^2 \leq \min(B, k/2 - |\rho_S|^2) + 2k_o|\rho_S| + k_o/2 \leq B + k_o\sqrt{2k} + k_o/2$, so $M(A(S \cup O))/\sqrt{|A|} \leq \frac{1}{2} + O(\sqrt{\log k/k}) + O(k_o/\sqrt{k}) \to \frac{1}{2}$ for $k_o = o(\sqrt{k})$, while the front-loading factor is of the order of the outliers' distance ratio. So the front-loading restriction is a limitation of the method, not of the sets.

Remark 7 (Localization is critical). Let $K \geq 0$ be a trigonometric polynomial with $\hat{K}(0) = 1$ and spectrum Σ, and let $\rho(\Sigma) = \sup\{\|h\|_2^2/\hat{h}(0)^2 : h \geq 0,\ \operatorname{spec} h \subset \Sigma\}$. Replacing h_{x_0} by $h_{x_0} * K$ in the proof of Theorem 2 yields

$$M^2\Psi(0) + \tfrac{1}{2}\sum_{a \in A}\Psi(a) \leq \rho(\Sigma)\Big(M^2 + \tfrac{1}{2}\sum_{a \leq w}\kappa_a^2 \mathbf{1}_A(a)\Big), \qquad \Psi = \hat{K}^2 * \kappa^2.$$

If A is full on the positive part of Σ, the two sides agree to leading order up to the factor $\|K\|_2^2/\rho(\Sigma) \leq 1$: no choice of K gives information. The global inequality gains the factor $2/\rho$ because the flat coefficient sequence of f_A is not itself a nonnegative kernel. (This is a leading-order computation, recorded to explain why the density threshold 0.4055 cannot be lowered by localizing the window.)

5. Formalization

The following are checked in Lean 4 with Mathlib (no `sorry`, only the axioms `propext`, `Classical.choice`, `Quot.sound`): the finite Fejér–Riesz theorem for every degree; the elementary Nikolskii constant of Lemma 1; Theorem 2 for 0/1 coefficients (`p31_elementary`); Corollary 5 with constant 1/1000 (`dense_elementary`); Theorem 2 for complex coefficients with $|b_j| \leq 1$ (`p34_elementary`); Corollary 3 in the form $-\min \geq \sqrt{n}/4$ for every $n \geq 1$ and all phases (`p34_A_phases`) and for ± 1 polynomials with both signs (`p34_B`); the dense complex-coefficient version (`p34_C`); and two auxiliary facts, $M(A) \geq |A| \min A/\max A$ and $|A \cap q\mathbb{N}| \geq |A|/q - (1 - 1/q)M(A)$.

References

[1] B. Bedert, *Polynomial bounds for the Chowla cosine problem*, arXiv:2509.05260.

[2] B. Bu et al., *A half-constant upper bound for Chowla's cosine problem*, arXiv:2610.17160v1 (2026).

[3] T. Erdélyi, *Bourgain–Ruzsa type estimates for the maximum and minimum of Littlewood cosine polynomials on the real line*, arXiv:2607.21877v1 (2026).

[4] A. Shankar, *A log-free $n^{1/5}$ bound for Chowla's cosine problem*, arXiv:2609.05338.